Daily Drilling Report
SPD19B-03
3/19/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
9.0	24"	15.00	72.83	31.70	90.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
26" CP, 178.00mKB	0.00	12,751.72	606,913.73		7.60
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
791.00	875.00		84.00	13.81	6.1

Operations at Report Time
POOH @ 140m.

Operations Summary
Drilling 24" hole f/ 791m to 875m (Section TD). Sweep hole. Perform wiper trip. Run back to 825m.

Operations next 24hrs
Cont' run back to Btm. Sweep hole. POOH Rotary BHA to surface. RIH 18 5/8" casing.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	3/31/2014
BOP Test		
Chemical Spill Drill		
Fire Drill	3/9/2014	3/11/2014
H2S Drill	3/16/2014	3/31/2014
Heli Crash Drill	2/20/2014	3/14/2014
Medical Drill		
MOB Drill	3/1/2014	3/23/2014
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports: 41

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	16:00	16....	24"	D	03	P	Drilling 24" hole f/ 791m to 875m (Section TD) w/ 1000 gpm / 1780 psi / 40-42 kips/ 55-60 rpm & 6 - 18 Kft-lbs. >> Drill ahead w/ complete loss, fill up annulus w/ SW. >>> Sweeping Sequence f/ 685m (Expected Top of Sachun Formation): ** At the Middle of each joint: Pump 40 bbl GG. ** At the End of each joint: Pump 40 bbl PHG. ** At the End of Stand: Pump 40 bbl GG + Circulate 5 min + 40 bbl PHG. B/R & R/D the stand. ** Take survey one single off bottom. ** Spot 30 bbl PHG + 20 bbl GG.
16:00	17:30	1.50	24"	D	07	P	Sweep hole by pumping 60 bbl GG + Circulate for 3 min + 100 bbl PHG w/ 1100 gpm/ 2100 psi/ 20 rpm while reciprocate the string.
17:30	18:00	0.50	24"	D	09	P	Spot first batch of 10 ppg weighted PHG (340 bbl) f/ TD to expected top of Sachun formation (685m) w/ 600 gpm/ 680 psi, while reciprocating string w/ 20 rpm.
18:00	18:30	0.50	24"	D	04	P	POOH Rotary BHA#3 f/ 875m to 685m. Hole in good condition.
18:30	19:00	0.50	24"	D	09	P	Spot second batch of 10 ppg weighted PHG (340 bbl) to loss zone (495m) w/ 600 gpm/ 620 psi/ 20 rpm.
19:00	20:00	1.00	24"	D	04	P	Cont' POOH f/ 685m to 325m.
20:00	21:30	1.50	24"	D	06	P	Had to B/R f/ 325m to 220m w/ 450-600 gpm/ 350-600 psi/ 20-35 rpm/ 7-24 kft.lbs
21:30	22:00	0.50	24"	D	04	P	Cont' POOH f/ 220m to 26" CP shoe @ 178m. Circulate w/ 1000 gpm/ 1800 psi and clean bit & stabilizer.
22:00	00:00	2.00	24"	D	04	P	Run back 24" rotary BHA f/ 178m to 825m in good hole condition.

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PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00		24"				Mar 20, 2014 00:00 - 01:00 Wash and R/D f/ 825m to 875m (Btm) w/ 400-600 gpm/ 500-750 psi/ 20-35 rpm/ 5-19 Kft-lbs. 01:00 - 02:30 Sweep hole by pumping 60 bbl GG + Circulate for 3 min + 100 bbl PHG w/ 1100 gpm/ 2100 psi/ 20 rpm while reciprocate the string. 02:30 - 03:00 Spot first batch of 10 ppg weighted PHG (340 bbl) f/ TD to expected top of Sachun formation (685m) w/ 600 gpm/ 680 psi, while reciprocating string w/ 20 rpm. 03:00 - 03:30 POOH Rotary BHA#3 f/ 875m to 685m. Hole in good condition. 03:30 - 04:00 Spot second batch of 10 ppg weighted PHG (340 bbl) to loss zone (495m) w/ 600 gpm/ 620 psi/ 20 rpm. 04:00 - 06:00 Cont' POOH f/ 685m to 140m.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
875.00	Sea Water	8.60				
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	5.	125	4.0	1,010.0	22.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Neptune-1	Crew Boat	3/20/2014 00:10	3/20/2014 00:45	Disembarked 21Pax and Embarked 14Pax for Kish.
DP3	Supply Boat	2/23/2014 23:30		Transferred DW-50MT. C/off and proceeded to anchorage.
Swissco Supplier	Supply Boat	3/19/2014 02:40		Delivered cargo & DW 200 MT , B/L for Kish &Sahar-1. Dep for Aban-8 at 1730 Hrs.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bentonite Bulk	6.5	105.0	64.5
Barite Bulk	15.7	221.0	166.0
Bio-Cide	7.0	139.0	65.0
Defoam SLC	2.0	139.0	59.0
Citric Acid	5.0	210.0	66.0
Caustic Soda	6.0	427.0	236.0
Guar Gam MP	50.0	1,421.0	961.0
Soda Ash	1.0	151.0	42.0
Lime	6.0	150.0	51.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	18.0	24.0	102.0
Cement G	0.0	0.0	183.0
Bentonite Bulk	6.5	0.0	56.5
Diesel Fuel	11.0	0.0	159.0
Drill Water	55.0	250.0	359.0
Barite BB	0.0	0.0	0.0
Bentonite BB	0.0	0.0	22.0
Barite Bulk	16.0	0.0	153.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
3	24	Varel	ETR14J	1250452	InserT Bit	435	24/24/24/20
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.60	64.92	230,247.00	491.00	875.00	384.00	5.9	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
850	1,100	1,250.0	2,100.0	22	40	40	80
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
83.22	64.92	100.00	0.00	280	256	219	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
791.00	834.00	8.89	16,000.0	1,000	55	1,780.0	40	255
834.00	875.00	4.92	17,000.0	1,000	60	1,780.0	40	256

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Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
13 Jts 5 7/8" HWDP	5 7/8	4.250	120.97	256.31
X/O	7	2.810	0.79	135.34
2 Jts 8 1/4" DC	8 1/4	2.875	18.75	134.55
8" Cougar DA Jar	8	3.500	5.84	115.80
5 Jts 8 1/4" DC	8 1/4	2.875	47.24	109.96
X/O	9 5/8	3.000	1.22	62.72
3 Jts 9 1/2" DC	9 1/2	3.000	28.29	61.50
17 1/2" Stabilizer	17 1/2	3.000	2.47	33.21
9 1/2" NMDC	9 1/2	4.000	9.45	30.74
9 1/2" TM Sub (MWD & Gamma probes inside)	9 1/2	3.125	2.15	21.29
9 1/2" NMDC	9 1/2	4.000	9.45	19.14
9 1/2" Restrictor sub (32/32" restrictor)	9 1/2	4.250	0.91	9.69
23" Stabilizer	23	3.000	2.86	8.78
9 1/2" Float Sub (Float valve inside)	9 1/2		0.79	5.92
9 1/8" Shock Sub	10 1/8	2.625	4.51	5.13
24" Insert Bit	24		0.62	0.62

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
781.70	249.40	20.40	760.92	-78.66	-121.63	144.79	0.36	0.31
810.40	249.80	21.10	787.75	-82.20	-131.16	154.77	0.75	0.73
839.70	249.60	21.40	815.06	-85.88	-141.12	165.20	0.32	0.31

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Asmari		374.50	372.44
Jahrum		493.00	487.29
Ilam		847.00	

General Remarks

Comment

>>> Requirements: (Shock Sub) + (300 MT DW) + (PDF MR#6 & 7).+ (100 MT FO).

>>> Sent:
*** Swissco Supplier to Rig Sahar-1: (Aban 350 T Manual Elevator).

*** Swissco Supplier to Kish: (COR 23" Stabilizer + Shock Sub) + (PPL Basket w/ casing protector) + (2 x Empty Food Container) + (Garbage).

>>> Received:
*** Swissco Supplier f/ Kish: (COR 23" Stabilizer) + (COR Fishing Tools: 12 3/4" FS Overshot + 11 1/4", 11", 10 7/8" Spiral Grapple + 10 1/4", 10", 9 3/4", 9 5/8", 9 1/2", 9 1/4' & 9" Basket Grapple + 17 1/2" Junk Mill + spear Stop Sub for 18 5/8" Casing + 20 1/2" Stop Spear Ring + 21" Oversize Lip Guide + 20" impression Block) + (Mehran CMT Additive).